Example 8.4 Find the sum of the deviations from the arithmetic mean for the following observations:

21, 30, 22, 16, 24, 28, 18, 17

Solution

$$\bar{X} = \frac{\sum_{i=1}^{8} x_i}{n} = \frac{21+30+22+16+24+28+18+17}{8} = \frac{176}{8} = 22$$

Deviation of an entry x_i from the arithmetic mean $\bar{X}$ is $x_i - \bar{X}$, $i = 1, 2, \ldots 8$.

Sum of the deviations

$= (21-22)+(30-22)+(22-22)+(16-22)+(24-22)+(28-22)+(18-22)+(17-22)$

$= 16-16 = 0.$ or equivalently, $\sum_{i=1}^{8}(x_i - \bar{X}) = 0$

Hence, we conclude that sum of the deviations from the Arithmetic Mean is zero.

Example 8.5 The arithmetic mean of 6 values is 45 and if each value is increased by 4, then find the arithmetic mean of new set of values.

Solution

Let $x_1, x_2, x_3, x_4, x_5, x_6$ be the given set of values then $\frac{\sum_{i=1}^{6} x_i}{6} = 45.$

If each value is increased by 4, then the mean of new set of values is

New A.M. $\bar{X} = \frac{\sum_{i=1}^{6}(x_i + 4)}{6}$

$$= \frac{(x_1+4)+(x_2+4)+(x_3+4)+(x_4+4)+(x_5+4)+(x_6+4)}{6}$$

$$= \frac{\sum_{i=1}^{6} x_i + 24}{6} = \frac{\sum_{i=1}^{6} x_i}{6} + 4$$

$\bar{X} = 45 + 4 = 49.$

Progress Check

Mean of 10 observations is 48 and 7 is subtracted to each observation, then mean of new observation is _____

Example 8.6 If the arithmetic mean of 7 values is 30 and if each value is divided by 3, then find the arithmetic mean of new set of values

Solution

Let X represent the set of seven values $x_1, x_2, x_3, x_4, x_5, x_6, x_7$.

Then $\bar{X} = \frac{\sum_{i=1}^{7} x_i}{7} = 30$ or $\sum_{i=1}^{7} x_i = 210$

If each value is divided by 3, then the mean of new set of values is

Progress Check

1. The Mean of 12 numbers is 20. If each number is multiplied by 6, then the new mean is __
2. The Mean of 30 numbers is 16. If each number is divided by 4, then the new mean is___

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